

Notes: Bloom and Van Reenen (QJE, 2007)

Measuring and Explaining Management Practices across Firms and Countries

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1 Big picture

- **Question.** Can management practices be measured in a systematic, comparable way across firms and countries, and do these practices help explain differences in productivity and other firm outcomes?
- **Setting.** The paper studies medium-sized manufacturing firms in the United States, the United Kingdom, France, and Germany using an original management survey.
- **Core challenge.** “Management quality” is usually left in the residual as a firm fixed effect. The paper tries to turn that residual into an observable object.
- **Main empirical innovation.** The authors build a survey-based measure of management using **18 practices**, scored from **1 (worst)** to **5 (best)**, and grouped into **operations, monitoring, targets, and incentives**.
- **Main validation result.** Better management is strongly associated with higher productivity, profitability, Tobin’s Q, sales growth, and survival.
- **Main descriptive result.** The United States has the highest average management scores, but the striking fact is the *large within-country dispersion*. A long left tail of badly managed firms drives much of the cross-country gap.
- **Main explanatory result.** Two forces help explain poor management: **low product market competition** and **family firms that pass control to the eldest son (primogeniture)**.
- **Quantitative result.** Competition and primogeniture together explain about **half of the tail of badly managed firms**, about **57%** of the U.S.–France gap, and about **29%** of the U.S.–UK gap in management scores.
- **Economic takeaway.** Management appears to be a form of organizational capital. It is not only about CEO talent or worker ability; it is embedded in the way firms monitor, target, reward, and improve operations.

2 Data and measurement (key objects)

2.1 Observed survey data and sample design

- The core dataset is an **original telephone survey of 732 medium-sized manufacturing firms**.
- Countries in the sample are:
 - United States: **290 firms**
 - United Kingdom: **151 firms**
 - France: **135 firms**
 - Germany: **156 firms**
- Firms are medium-sized, with employment between **50 and 10,000 workers**; the median firm has about **675 employees**.
- The survey achieved a relatively high **54% response rate**.

- The sampling frame was designed to be representative of medium-sized manufacturing firms in each country. The order in which firms were contacted was randomized.
- Interviews were conducted by specially trained MBA or graduate-student interviewers, in the manager’s native language, and each interviewer completed many interviews so that interviewer fixed effects could later be controlled for.

2.2 Management score construction

- Each firm is scored on **18 practices**, each from **1 to 5**.
- These 18 practices are grouped into four areas:
 - **Operations** (3 practices)
 - **Monitoring** (5 practices)
 - **Targets** (5 practices)
 - **Incentives** (5 practices)
- Examples include lean manufacturing, process documentation, performance tracking, performance review, target realism, promotion criteria, and firing of poor performers.
- The empirical measure used in most regressions is the average of practice-level **z-scores**:

$$M_i = \frac{1}{18} \sum_{p=1}^{18} \frac{s_{ip} - \bar{s}_p}{\sigma_p},$$

where s_{ip} is firm i ’s score on practice p .

- Standardizing practice by practice is useful because some questions are naturally more compressed or more dispersed than others.

2.3 Additional firm-level and industry-level data

- The survey is linked to accounting and market data on:
 - sales,
 - employment,
 - capital,
 - materials,
 - profits,
 - Tobin’s Q,
 - wages when available,
 - exit,
 - sales growth.
- The authors also collected a separate human-resources survey with worker characteristics such as education, hours, age, and organizational structure.

- Family ownership, family management, and succession rules were assembled from public data and follow-up surveys.
- Competition is measured three ways:
 - lagged **import penetration**,
 - a lagged **Lerner-type competition measure**,
 - managers' **self-reported number of competitors**.

2.4 Internal validation and measurement error

- The authors re-interviewed **64 firms**, typically using a different manager and a different interviewer.
- The correlation between first and second interview scores is **0.734**, which is strong evidence that the survey captures real firm-level practices rather than random noise.
- Measurement error is smaller in the overall firm average than in individual practices: the paper estimates that about **25%** of the variation in the firm-level average score is measurement error, versus about **42%** at the individual-practice level.
- Responding firms look similar to nonresponders on most observables, except that responders are somewhat larger on average.

2.5 Key descriptive objects

- Average raw management scores are:
 - U.S.: **3.32**
 - Germany: **3.27**
 - France: **3.11**
 - U.K.: **3.04**
- Decomposition of variation in management scores:
 - only about **2%** is across countries,
 - about **42%** is across country-by-industry cells,
 - about **56%** is within country and industry.
- Family ownership is much more common in Europe than in the United States. Roughly **30%** of European firms have a family as the largest shareholder, compared with about **10%** in the U.S.
- Primogeniture family succession is especially common in the U.K. and France: around **14–15%** of all firms there are family-controlled with family CEO succession by eldest son, compared with only about **3%** in the U.S. and Germany.

3 Models and empirical specifications (all equations + interpretation)

3.1 Conceptual framework: why management practices differ

The paper begins with a conceptual discussion rather than a formal structural model.

- **Good management may not be universal.** Some practices depend on firm environment, worker skill, and technology.
- **Static reasons for heterogeneity.**
 - Different management practices have different costs.
 - Agency problems may make managers prefer “bad” practices if better practices require more effort or transparency.
 - Firms differ in their optimal practices because technologies and workforces differ.
- **Dynamic reasons for heterogeneity.**
 - Firms may learn slowly about new practices.
 - Adoption may be costly and gradual.
 - Selection may matter: competition may remove poorly managed firms over time.

Interpretation. The paper is not saying every practice is always universally optimal. It argues that a substantial subset of practices can plausibly be judged as better or worse, and these measurable practices are empirically linked to performance.

3.2 Productivity equation

The core validation regression is a firm production function:

$$y_{itc} = \alpha_l^c l_{itc} + \alpha_k^c k_{itc} + \alpha_n^c n_{itc} + \beta^c M_i + \gamma^c Z_{itc} + u_{itc}.$$

Here:

- y_{itc} is log deflated sales,
- l_{itc} is log labor,
- k_{itc} is log capital,
- n_{itc} is log intermediate inputs (materials),
- M_i is the management score,
- Z_{itc} includes controls such as worker education, hours, firm age, listing status, industry dummies, and country effects.

Interpretation. The coefficient β measures whether better managed firms have higher productivity, conditional on standard inputs and controls.

3.3 Alternative performance regressions

The paper then examines whether the management score predicts other dimensions of firm performance:

$$\begin{aligned}\text{ROCE}_i &= \beta M_i + \gamma X_i + \varepsilon_i, \\ \ln(Q_i) &= \beta M_i + \gamma X_i + \varepsilon_i, \\ \text{Pr}(\text{Exit}_i = 1) &= \Phi(\beta M_i + \gamma X_i), \\ \text{Sales Growth}_i &= \beta M_i + \gamma X_i + \varepsilon_i.\end{aligned}$$

Interpretation. If the management score is meaningful, it should correlate not only with productivity but also with profitability, valuation, growth, and survival.

3.4 Econometric approaches

- **Cross-sectional OLS.** The simplest specification relates outcomes directly to the management score and controls.
- **Olley–Pakes two-step approach.** The paper also estimates a productivity specification allowing the unobserved firm efficiency term to follow a Markov process.
- **Within-groups and GMM variants.** These are used as robustness checks to address input endogeneity and persistent unobservables.

Interpretation. The authors are not claiming a fully clean causal estimate of management on productivity. The point is that the management measure survives a battery of standard productivity specifications and therefore passes a strong external-validity test.

3.5 Skill-contingent management practices

The paper distinguishes between management practices that are more “human-capital focused” and those that are more “fixed-capital focused.”

A generic specification is

$$M_i^H - M_i^K = \delta \cdot \text{Skill}_i + \gamma X_i + \varepsilon_i,$$

where M_i^H is human-capital management, M_i^K is fixed-capital management, and Skill_i is measured using worker education or average wages.

Interpretation. This section shows that management is not one-dimensional. Higher-skill firms tend to adopt more human-capital-intensive management practices.

3.6 Competition regressions

To study whether competition is associated with better management, the paper estimates regressions of the form

$$M_i = \delta \cdot \text{Competition}_i + \gamma X_i + \varepsilon_i.$$

The main competition measures are:

- lagged import penetration,
- a lagged industry Lerner-style competition index,
- a self-reported competitor count coded as 0 (none), 1 (less than 5), and 2 (5 or more).

Interpretation. Tougher product market competition is associated with better management. This can arise either because competition disciplines incumbents or because badly managed firms are selected out.

3.7 Family-firm regressions

The paper estimates management-score regressions using indicators for family ownership and family management:

$$M_i = \beta_1 \text{FamilyOwner}_i + \beta_2 \text{FamilyCEO}_i + \beta_3 \text{Primogeniture}_i + \gamma X_i + \varepsilon_i.$$

Definitions:

- **Family largest shareholder:** a single family is the largest ownership block.
- **Family largest shareholder and family CEO:** a family member is also CEO.
- **Family largest shareholder, family CEO, and primogeniture:** the CEO was selected as the eldest male child.

Interpretation. The paper’s key message is that family ownership itself is not the main problem. The negative effect is concentrated in firms that combine family control with *primogeniture* succession.

3.8 Cross-country decomposition

Finally, the paper asks how much of the U.S.–Europe management gap can be accounted for by competition, primogeniture, and skill:

$$M_i = \theta_{GER} D_{GER,i} + \theta_{FRA} D_{FRA,i} + \theta_{UK} D_{UK,i} + \phi \text{Primogeniture}_i + \lambda \text{Competition}_i + \psi \text{Skill}_i + \gamma X_i + \varepsilon_i.$$

Interpretation. This is not a decomposition in a strict causal sense, but it shows how much observed cross-country differences shrink once these covariates are added.

4 Identification and empirical credibility (what makes the findings believable)

4.1 Why measurement is difficult

- Management is often treated as an unobserved firm fixed effect.
- Direct surveys can suffer from interviewer bias, respondent self-presentation bias, and sampling bias.
- Cross-country comparisons are especially hard because “good management” could reflect an Anglo-Saxon coding scheme rather than real differences.

4.2 Survey design choices that improve credibility

- **Double blind design.** Interviewers did not know firm performance, and managers did not know they were being scored.
- **Open-ended questions.** Managers were asked to describe concrete practices instead of choosing from a menu of “good” or “bad” answers.
- **Telephone interviews in native language.** This reduces some cross-country communication problems.
- **Interviewer controls.** Because each interviewer conducted many interviews, interviewer fixed effects can be included in regressions.
- **Noise controls.** The paper controls for characteristics of the interview process itself: time of day, day of week, interview duration, manager tenure/seniority, interviewer identity, and a reliability score.

4.3 Internal and external validation

- **Internal validation.** Repeat interviews on the same firm yield a correlation of 0.734.
- **External validation.** The management score predicts productivity, ROCE, Tobin’s Q, sales growth, and survival using independent accounting and market data.

4.4 What is and is not identified

- The paper identifies strong *associations* between management and performance.
- It does *not* claim a clean causal estimate of management on productivity from the production-function regressions.
- Likewise, the effects of competition and primogeniture are mainly conditional correlations, although the authors argue that several likely biases would attenuate the estimated effects rather than exaggerate them.

4.5 Evidence against simple alternative stories

- **Not just technology.** Adding available measures of PCs per worker or R&D does little to weaken the management coefficient.
- **Not just higher worker quality.** Controlling for wages leaves the management coefficient largely intact, and CEO pay is not correlated with management conditional on firm size.
- **Not just short-run mood or optimism.** Management scores are not significantly related to lagged productivity growth, which weakens the “good times make managers sound optimistic” interpretation.

5 Tables and figures

5.1 Figure I: Distribution of management scores by country

Read:

- This is the paper’s most important descriptive figure.
- The U.S. distribution lies to the right of the European distributions, but the bigger fact is the *wide dispersion within each country*.
- A long left tail of badly managed firms drives much of the cross-country difference.
- Firms with scores of 2 or less are not just a little worse; they have very weak monitoring, targets, and incentives.

5.2 Table I: Estimates of firm performance equations

Read:

- The central productivity result is that management enters positively and significantly in the production function.
- In the preferred OLS productivity specification with full controls and interview-noise controls, the management coefficient is about **0.040** (s.e. **0.012**).
- In the Olley–Pakes specification, the coefficient is about **0.038** (s.e. **0.015**), and alternative GMM/within-groups versions are similarly positive.
- Management also predicts other outcomes:
 - **ROCE**: coefficient about **2.452**
 - **Tobin’s Q**: coefficient about **0.258**
 - **Sales growth**: coefficient about **0.019**
 - **Exit**: negative marginal effect, about **-0.200**
- So the management score is not just related to one accounting measure; it is broadly related to better firm outcomes.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Estimation method	OLS	OLS	OLS	OLS	Olley-Pakes	OLS	OLS	Probit	OLS
Firms	All	All	All	All	All	All	All	All	All
Dependent variable	Ln(T) _{it}	Ln(T) _{it}	Ln(T) _{it}	Ln(T) _{it}	Ln(T) _{it}	ROCE	Ln(Tobin's q) _{it}	Exit (by death)	Sales growth
	Sales	Sales	Sales	Sales	Sales	Profitability			
Management z-score	0.075 (0.024)	0.039 (0.012)	0.032 (0.011)	0.040 (0.012)	0.038 (0.015)	2.452 (0.876)	0.258 (0.072)	-0.200 (0.024)	0.019 (0.006)
Ln(L) _{it}	1.080 (0.034)	0.522 (0.036)	0.535 (0.033)	0.522 (0.032)	0.426 (0.022)	1.432 (1.712)	0.400 (0.194)	0.233 (0.043)	-0.021 (0.014)
Labor		0.186 (0.029)	0.147 (0.028)	0.147 (0.025)	0.158 (0.042)	-1.855 (1.390)	-0.080 (0.170)	0.009 (0.056)	0.009 (0.012)
Ln(K) _{it}		0.301 (0.037)	0.308 (0.028)	0.307 (0.025)	0.412 (0.040)	1.681 (1.025)	0.286 (0.110)	-0.084 (0.202)	0.008 (0.009)
Capital									
Ln(N) _{it}									
Materials									
Country, time, and industry dummies	Yes	Yes	Yes	Yes	Yes	Yes	Yes	No	Yes
General controls	No	No	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Noise controls	No	No	No	Yes	Yes	Yes	Yes	Yes	Yes
Firms	709	709	709	709	709	690	374	709	702
Observations	5,350	5,350	5,350	5,350	5,350	3,696	5,099	709	4,777

Notes: All columns estimated by OLS except column (8), which is estimated by probit maximum likelihood, and column (5), which is estimated using the Olley and Pakes (1996) technique. In all columns except (8), standard errors are in parentheses under coefficient estimates and allow for arbitrary heteroskedasticity and serial correlation (i.e., clustered by firm). In column (8), we report the p-value in square brackets below the marginal effects of each variable on the percentage increase in the probability of exit between 2004 and 2005. The coefficients on capital, materials, and labor are allowed to be different across countries and consolidation status (United Kingdom is base). "General controls" comprise firm-level controls for leverage, hours worked, leverage age, a dummy for being listed, a dummy for consolidated accounts, the share of the workforce with degrees, and the share of the workforce with MBAs (except column (8), which just controls for leverage and listing status). "Noise controls" are 16 interviewer dummies, the seniority, gender, tenure and number of countries worked in of the manager who reported age, the day of the week the interview was conducted, the time of the day the interview was conducted, the duration of the interview, and an indicator of the reliability of the information as coded by the interviewer. Data run between 1994 and 2004, except in column (8), which is a cross section. All regressions include a full set of three-digit industry dummies and four country dummies interacted with a full set of time dummies (except column (5), which has a linear time trend and country dummies, and column (8). Column (5) uses a third-order series expansion in ln(sales) and ln(investment), and we also include a selection correction term following Olley and Pakes (1996). Standard errors are bootstrapped (clustered by firm) with 200 replications. After calculating the parameters of labor and materials (stage 1a) and capital (stage 1b), we calculate the efficiency term/TFP averaged by firm across all years. This is used as a dependent variable and regressed on the management score and the general controls (stage 2).

	(1)	(2)	(3)	(4)	(5)
Dependent variable	Human capital management	Fixed capital management	Human capital – fixed capital management	Human capital – fixed capital management	Human capital – fixed capital management
Ln(proportion of employees with college degrees)	0.198 (0.043)	0.102 (0.047)	0.096 (0.049)	0.099 (0.057)	
Ln(firm average wages) _{it}					0.340 (0.168)
General controls	No	No	No	Yes	Yes
Industry controls	No	No	No	Yes	Yes
Firms/industries	732	732	732	732	424

Notes: All columns estimated by OLS with robust standard errors in parentheses. A single cross section of data is used. "Human capital management" is the average z-score of the three explicitly human-capital-focused practices (practices 13, 17, and 18 in Appendix I.A). "Fixed capital management" is the average z-score of the three most fixed-capital-focused practices (1, 2, and 4 in Appendix I.A). "Human capital – fixed capital management" is the difference of these two averages. "General controls" comprise controls for leverage age, ln(average number of employees), a dummy for being listed, and a set of country dummies. "Industry controls" are a full set of three-digit industry dummies.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Estimation method	OLS	OLS	OLS	OLS	OLS	OLS	OLS	OLS
Dependent variable	Management z-score	Management z-score	Management z-score	Management z-score	Management z-score	Management z-score	Management z-score	Management z-score
Import penetration (5-year lagged)	0.144 (0.045)	0.166 (0.071)					0.123 (0.044)	0.180 (0.073)
Lerner index (5-year lagged)			1.516 (0.494)	1.192 (0.568)			1.204 (0.621)	1.257 (0.592)
Number of competitors					0.143 (0.051)	0.140 (0.040)	0.125 (0.043)	0.120 (0.038)
Firms	732	732	726	726	722	722	726	726
General controls	No	Yes	No	Yes	No	Yes	No	Yes

Notes: Coefficients from OLS regressions with standard errors in parentheses (robust to arbitrary heteroskedasticity and clustered by country $\times$ industry pair). Sample is a single cross section. General controls includes a full set of three-digit industry dummies, four country dummies, ln(firm size), ln(firm age), a dummy for being listed, the share of workforce with degrees, the share of workforce with MBAs, a dummy for being consolidated, and the nine controls (16 interviewer dummies, the seniority, gender, tenure, and number of countries worked in of the manager who responded, the day of the week the interview was conducted, the time of the day the interview was conducted, the duration of the interview, and an indicator of the reliability of the information as coded by the interviewers). Import penetration = $\ln(\text{import}/\text{production})$ in every country $\times$ industry pair with the average over 1990–1999 used. Lerner index of competition is constructed, as in Aghion et al. (2005), as the mean of $(1 - \text{profit}/\text{sales})$ in the entire database (excluding the firm itself) for every country-industry pair coverage over 1990–1999 used. Number of competitors is constructed from the response to the survey question on number of competitors, and is coded as zero for none (1% of responses), 1 for less than 5 (51% of responses), and 2 for “5 or more” (49% of responses).

5.3 Table II: Skill-contingent management practices

Read:

- Higher-skill firms use relatively more *human-capital-focused* management practices than fixed-capital-focused ones.
- The share of employees with college degrees is more strongly associated with human-capital management than with fixed-capital management.
- The same pattern appears when skill is proxied by average wages.
- This matters because it shows the paper is not claiming one universal template of management; some practices are clearly contingent on workforce quality.

5.4 Table III: Management and product market competition

Read:

- All three competition measures point the same way: more competition is associated with better management.
- A particularly intuitive summary is the self-reported competition result: moving from “few” competitors to “many” competitors raises the management z-score by about **0.140**.
- The result survives controls for country, industry, firm size, age, listing, and interview noise.
- The paper argues that if anything, endogeneity is likely to attenuate this result because good management can itself raise profits and distort some competition measures.

5.5 Table IV and Table V: Family ownership and management

Read:

- Table IV shows that family involvement is common, especially in Europe.
- Table V shows that *family ownership alone* is not the main problem.
- The really negative coefficient is for firms with family ownership, family CEO, and **primogeniture**. In the controlled specification the coefficient is about **-0.317**; without the general controls it is even more negative (about **-0.590**).
- When all family-firm indicators are entered together, family ownership by itself is mildly positive or insignificant, while primogeniture remains strongly negative.

TABLE IV
HEREDITARY FAMILY FIRM INVOLVEMENT BY COUNTRY

%	France	Germany	UK	U.S.
Family largest shareholder	30	32	31	10
(of which) Family largest shareholder and family CEO	19	11	23	7
(of which) Family largest shareholder, family CEO, and primogeniture	14	3	15	3
Founder largest shareholder	26	5	15	18
(of which) Founder largest shareholder and CEO	19	1	12	11
Number of firms	125	152	150	290

Notes. These mean values are taken from our sample of 717 firms. Family shareholding is combined across all family members. Family involvement is defined as second-generation family or beyond. Primogeniture is defined by a positive answer to the question "How was the management of the firm passed down: was it to the eldest son or by some other way?" Alternatives to primogeniture in frequency order are younger sons, sons-in-law, daughters, brothers, wives, and nephews. "Family largest shareholder" firms defined as those with a single family (combined across all family members, who are all second generation or beyond) as the largest shareholder; "family largest shareholder and family CEO" firms are those with additionally a family member as the CEO; "family largest shareholder, family CEO, and primogeniture" with additionally the CEO selected as the eldest male child upon succession. See Appendix II for more details on construction of the variables.

TABLE V
MANAGEMENT AND FAMILY FIRMS

	(1)	(2)	(3)	(4)	(5)	(6)
Estimation method	OLS	OLS	OLS	OLS	OLS	OLS
Sample	All	All	All	All	All	Family and external owners Management
Dependent variable	Management z-score	Management z-score	Management z-score	Management z-score	Management z-score	Management z-score
Family largest shareholder	0.005 (0.063)				0.138 (0.086)	0.137 (0.090)
Family largest shareholder and family CEO		-0.105 (0.075)			-0.010 (0.113)	-0.040 (0.114)
Family largest shareholder, family CEO, and primogeniture			-0.317 (0.096)	-0.590 (0.098)	-0.410 (0.122)	-0.379 (0.128)
Firms	732	732	732	732	732	618
Country controls	Yes	Yes	Yes	Yes	Yes	Yes
General controls	Yes	Yes	Yes	No	Yes	Yes

Notes. Coefficients from OLS regressions with standard errors in parentheses (robust to arbitrary heteroscedasticity). The sample is a single cross section. In columns (1) to (5), the complete sample is used; in column (6), founder firms are dropped. "General controls" are a full set of three-digit industry dummies, ln(firm size), ln(firm age), a dummy for being listed, share of workforce with degrees, share of workforce with MBAs, a dummy for being consolidated, and the noise controls (16 interviewer dummies, the sector, gender, tenure, and number of countries worked in of the manager who responded, the day of the week the interview was conducted, the time of day the interview was conducted, the duration of the interview, and an indicator of the reliability of the information as coded by the interviewer).

- So the paper's message is precise: heredity-based succession appears harmful, not family ownership in general.

5.6 Figure II: Tail of badly managed firms

Read:

- Figure II splits firms into two groups:
 - low competition and/or primogeniture,
 - high competition and no primogeniture.
- The badly managed tail is much thicker in the first group.
- In the low-competition/primogeniture group, **9.7%** of firms score 2 or less.
- In the high-competition/nonprimogeniture group, only **2.9%** do.
- This is the clearest visual evidence that these two factors explain a large share of extreme bad management.

5.7 Table VI: Accounting for management practices across countries

Read:

- Relative to the U.S., raw gaps in management are about:
 - Germany: **-0.045**
 - France: **-0.202**
 - U.K.: **-0.276**
- Adding the primogeniture variable shrinks the France and U.K. gaps.
- Adding competition shrinks them further, especially for France.
- Competition and family structure explain about **57%** of the U.S.–France gap and about **29%** of the U.S.–U.K. gap.

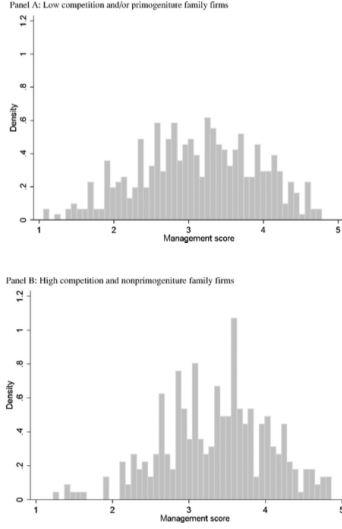


FIGURE II

The Distribution of Management Scores Split by Production Market Competition and Family Firms

Notes: Panel A shows average management scores for the 414 firms which (i) report facing “few” or “no” competitors, and/or (ii) have a family (second generation or more) as the largest shareholder with a family CEO chosen by primogeniture. Split by country is France (95), Germany (101), UK (84) and the U.S. (134). Overall 9.7% of the sample score two or less. 1 indicates worst practice, 5 indicates best practice. Panel B shows average management scores for the 308 firms which report facing “many” competitors and do not have a family (second generation or more) as the largest shareholder with a family CEO chosen by primogeniture. Split by country is France (34), Germany (51), UK (67) and the U.S. (156). Overall 2.9% of the sample score two or less. 1 indicates worst practice, 5 indicates best practice.

	(1)	(2)	(3)	(4)	(5)
Estimation method	OLS	OLS	OLS	OLS	OLS
Dependent variable	Management raw score	Management raw score	Management raw score	Management raw score	Management raw score
Country is the United States	Baseline	Baseline	Baseline	Baseline	Baseline
Country is Germany	-0.045 (0.064)	-0.081 (0.075)	-0.096 (0.075)	-0.057 (0.074)	0.004 (0.077)
Country is France	-0.202 (0.086)	-0.183 (0.104)	-0.136 (0.104)	-0.078 (0.103)	-0.033 (0.103)
Country is the United Kingdom	-0.276 (0.078)	-0.276 (0.093)	-0.227 (0.091)	-0.196 (0.091)	-0.123 (0.093)
Family largest shareholder, family CEO, and primogeniture			-0.637 (0.101)	-0.627 (0.100)	-0.582 (0.098)
Number of competitors				0.149 (0.052)	0.158 (0.051)
Ln(proportion of employees with degrees)					0.146 (0.037)
Controls for size and listed status	No	Yes	Yes	Yes	Yes
Firms	732	732	732	732	732

Notes: Coefficients from OLS regressions with standard errors in parentheses (robust to arbitrary heteroscedasticity). The sample is a single cross section. “Family largest shareholder, family CEO, and primogeniture” is a binary indicator for whether the family is the largest shareholder and the CEO is chosen by primogeniture. “Number of competitors” is constructed from the response to the survey question on number of competitors and is coded as zero for “none” (1% of responses), 1 for “less than 5” (31% of responses), and 2 for “5 or more” (68% of responses).

- Adding worker skill (college share) explains additional remaining differences.
- This table is the paper’s attempt to move from “correlates” to a simple quantitative accounting exercise.

6 Short summary

- **Method.** The paper builds a new management dataset by scoring firms on 18 practices using a double-blind survey design and linking those scores to firm accounts and ownership data.
- **Validation.** The management score is internally reliable and externally predictive of productivity, profitability, market valuation, sales growth, and survival.
- **Main findings.** The United States is better managed on average than the U.K. and France, but the dominant fact is large within-country heterogeneity and a long left tail of badly managed firms.
- **Mechanisms.** Low product market competition and family succession by primogeniture are strongly associated with poor management and explain a large share of both the left tail and cross-country gaps.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

- Management practices can be measured in a systematic, comparable way across firms and countries.
- These measured practices are not just survey noise; they are economically meaningful and strongly related to firm performance.
- The biggest management differences are not only between countries but within them, with a striking tail of badly managed firms.

7.2 Economic intuition

- Competition limits the survival of bad practices and may also sharpen managerial incentives.
- Family ownership is not necessarily harmful, but hereditary succession to management can reduce talent selection and weaken organizational discipline.
- Good management looks like organizational capital: it is embedded in routines, monitoring systems, targets, and personnel policies, not just in one talented manager.

7.3 Takeaway

This paper turns “management quality” from an unobserved residual into a measurable economic object. It then shows that management matters for firm performance and that institutions such as competition and succession rules help explain why badly managed firms persist. In that sense, the paper connects productivity, organization, corporate governance, and comparative economic performance in a single empirical framework.